SLIC Superpixel Segmentation for Polarimetric SAR Images

Junjun Yin, *Senior Member, IEEE*, Tao Wang, Yanlei Du, *Member, IEEE*, Xiyun Liu, Liangjiang Zhou, and Jian Yang, *Senior Member, IEEE*

***Abstract*— Superpixel segmentation approaches for polarimetric synthetic aperture radar (SAR) images have only been studied in recent years. Simple linear iterative clustering (SLIC) is a simple and efficient superpixel segmentation method, first proposed for optical images. It basically includes three implementation steps, i.e., initialization, local k-means clustering, and postprocessing. The challenge of applying SLIC to polarimetric SAR images lies in constructing the effective spatial and feature similarity and proposing the efficient segmentation procedure. In this study, to address both issues, we modify the SLIC clustering function to adapt the characteristics of polarimetric statistical measures. A new initialization method is proposed, which exploits the image gradient information to produce robust cluster centers. Furthermore, in an effort to give a comprehensive comparison and provide a fair assessment of the feature similarities for polarimetric SAR imagery, four classic statistical distances, among which two were not studied along with the SLIC previously, are embedded in the modified clustering function. The proposed method is validated by comparing with state-of-the-art SLIC-based algorithms and also the Ncut and TurboPixel algorithms. Experiments on extensive polarimetric SAR data sets show that the proposed method can significantly improve the segmentation results, with producing better boundary adherence and compact as well as uniform superpixels. We also obtain distinct conclusions that are different from the existing studies when investigating the performances of the statistical measures.**

***Index Terms*— Radar polarimetry, simple linear iterative clustering (SLIC), statistical measure, superpixel, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a high-resolution active microwave remote sensing imaging system that provides an all-day and all-weather means of Earth observation [1]. Conventional SAR uses a single-polarized transmitting and receiving antenna such that only one component of the scattering matrix can be obtained. Using a pair of orthogonal polarizations for alternatively measuring the backscattered wave, the polarimetric SAR can obtain more information to represent the scattering characteristics of ground targets. This significantly improves the SAR remote sensing applications, such as image classification, segmentation, target detection, and recognition [2]. In remote sensing images, adjacent pixels belonging to the same ground object typically contain similar information. If these pixels are treated individually in the image processing, such as the pixel-based classification, the spatial contextual information is omitted. In order to take advantage of the pixel neighborhood information, one of the techniques is superpixel segmentation [3], [4]. The superpixel technique was first proposed by Ren and Malik [5] for optical images to divide the image into a set of uniform blocks composed of pixels with high similarities in shape, color, texture, and so on. The uniform block was named the superpixel. By segmenting the image into hundreds of superpixels, visual tasks, such as the object class recognition and image segmentation, can benefit from the oversegmentation.

Over the last decade, various superpixel algorithms have been proposed [6]. The commonly used methods for superpixel generation are based on the graph theory and the clustering technique. Methods based on the graph theory include the entropy-based algorithm [7], the Ncut algorithm [8], and the graph-based algorithm [9]. Clustering-based superpixel methods include the simple linear iterative clustering (SLIC) method [10], the mean-shift method [11], the iterative edge refinement (IER) method [12], and the watershed segmentation [13]. Those methods have the merits of either fast processing speed or good boundary fitting degrees, but sometimes also have the drawbacks of generating superpixels with very irregular shapes and sizes. Recently, the SLIC algorithm has been widely used due to its simplicity, effectiveness, and computational efficiency [6]. The SLIC segmentation framework consists of three steps: initialization, local k-means clustering, and postprocessing. However, when applied to the multichannel SAR images, SLIC shows a poor performance in generating superpixels due to the inherent speckle noise. Several SLIC-based algorithms have been developed for polarimetric SAR imagery. Feng *et al.* [14] used the Wishart distance instead of the Euclidean distance to measure the similarity of backscatter in the feature space. Xie *et al.* [15] adopted another distance, which is a combination of the spatial, Wishart, and edge-gradient distances. Qin *et al.* [16] investigated the initialization and postprocessing steps and improved the SLIC algorithm based on the

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revised Wishart (RW) distance. This method was named grid-centered initialization-based SLIC (GC-SLIC), which can obtain a reasonable superpixel boundary fitting degree. Zhang *et al.* [17] applied the IER algorithm to polarimetric SAR images and significantly improved the computational efficiency. Xiang *et al.* [19] adopted the distance measure based on the statistical model of the spherical invariant random vector (SIRV), which incorporates a homogeneity measure defined by the equivalent number of looks (ENL) in addition to the spatial pixel distance [18].

The segmentation performance is affected by both the segmentation procedure and the distance (or similarity) measure. In previous studies, the SLIC-based algorithms for polarimetric SAR images mainly focused on the improvement of establishing an effective distance measure to discriminate between classes. However, some issues were still not well addressed. First, in the segmentation procedure, the initialization step is crucial for the resulted accuracy. When a random initialization is used for SLIC, the class centers may converge slowly and oscillate among the neighboring superpixels. However, a few initialization methods have ever been studied for the polarimetric SAR SLIC approach. Second, in the iteration step, the clustering function (or distance measure) adopted the same form as that used in the optical images (see [10, eq. (2)]), which is not appropriate for some of the polarimetric statistical measures, e.g., the classic Wishart distance. However, some studies did not notice this. Furthermore, only the Wishart distance and the RW distance were studied for SLIC, but there are some other statistical measures that were developed for other applications, such as the Hotelling–Lawley trace (HLT) [20] for change detection and the symmetric revised Wishart (SRW) distance for spectral partitioning, which can also be embedded in the SLIC framework but have not been investigated.

This article studies the SLIC approach by considering both its adaption to various polarimetric similarity measures and the effect of initial center seeds. The contributions are twofold.

On the aspect of the segmentation procedure, first, the combined form of the statistical distance and the spatial proximity is modified, which is used as the SLIC clustering function. By this means, different statistical measures can be unified in the same segmentation framework such that the polarimetric statistical measures can be compared fairly. Second, a new initialization method is proposed, which exploits the polarimetric SAR edge information to provide stable center seeds.

On the aspect of the statistical measure, we review and analyze the performances of four commonly used test statistics for polarimetric SAR images to give a comprehensive comparison. Not only are the distances that have already been applied to SLIC included but other classic statistical measures that have not been embedded in the SLIC are also studied.

The rest of this study is organized as follows. Section II introduces the statistical measures for multilook polarimetric SAR data. Section III introduces the SLIC segmentation framework, where the proposed initialization method and the modified clustering function are presented. Experiments and comparisons are given in Section IV. Conclusions are drawn in Section V.

II. Statistical Distances

The quad-polarimetric SAR measures the backscattered signals from targets with four polarimetric components, which can be described by the complex scattering matrix $\boldsymbol{S}$

$$\boldsymbol{S} = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{1}$$

where $S_{\mathrm{HV}} = S_{\mathrm{VH}}$ for a reciprocal medium. The Pauli-basis scattering vector $\boldsymbol{k}_{\mathrm{P}}$ is defined as

$$\boldsymbol{k}_{\mathrm{P}} = \frac{1}{\sqrt{2}}[S_{\mathrm{HH}} + S_{\mathrm{VV}} \;\; S_{\mathrm{HH}} - S_{\mathrm{VV}} \;\; 2S_{\mathrm{HV}}]^{\mathrm{T}} \tag{2}$$

where the superscript $^{\mathrm{T}}$ denotes matrix transpose. For multilook polarimetric SAR data, each pixel can be represented by a coherency matrix $\boldsymbol{T}$, which is Hermitian positive definite (HPD), defined as

$$\boldsymbol{T} = \frac{1}{L}\sum_{i=1}^{L} \boldsymbol{k}_{Pi}\boldsymbol{k}_{Pi}^{*\mathrm{T}} \tag{3}$$

where the superscript $^{*\mathrm{T}}$ denotes the conjugate transpose of a matrix and L is the number of looks.

Assume that elements of the scattering matrix follow the complex multivariate Gaussian distribution with zero mean. Then, the coherency matrix $\boldsymbol{T}$ can be modeled by the complex Wishart distribution. The probability distribution function (PDF) is given by the following expression [21]:

$$p(\boldsymbol{T}|L, \boldsymbol{\Sigma}) = \frac{L^{qL} \;|\boldsymbol{T}|^{L-q} \exp\left(-L\,\mathrm{Tr}[\boldsymbol{\Sigma}^{-1}\boldsymbol{T}]\right)}{\mathrm{K}(L,q)\;|\boldsymbol{\Sigma}|^{L}} \tag{4}$$

with

$$\mathrm{K}(L,q) = \pi^{q(q-1)/2}\prod_{j=1}^{q}\Gamma(L-j+1) \tag{5}$$

where $\boldsymbol{\Sigma} = \mathrm{E}\left(\boldsymbol{k}\boldsymbol{k}^{*\mathrm{T}}\right)$ and q denotes the dimension of the vector $\boldsymbol{k}$. For quad-pol data in the backscatter (monostatic) case, $q = 3$, $\Gamma(\cdot)$ is the Gamma function, and $\mathrm{Tr}(\cdot)$ represents the trace operation to a matrix. In Sections II-A–II-D, we will introduce four classic dissimilarity statistical distances that have been successfully used for polarimetric SAR applications.

A. Wishart Distance

The Wishart distance is the most widely used statistical distance for polarimetric SAR data, which is derived based on the logarithm of the probability distribution of observations. The distance represents the likelihood of a pixel given an image model, which is defined as follows:

$$\begin{aligned} D_{\mathrm{W}}(i,j) &= -\frac{1}{L}\ln p(\boldsymbol{T}_i|L, \hat{\boldsymbol{\Sigma}}_j) - c \\ &= \ln(|\hat{\boldsymbol{\Sigma}}_j|) + \mathrm{Tr}\left(\hat{\boldsymbol{\Sigma}}_j^{-1}\boldsymbol{T}_i\right) \end{aligned} \tag{6}$$

where $\hat{\boldsymbol{\Sigma}}_j$ is the mean coherency matrix of the jth class, $\boldsymbol{T}_i$ is the coherency matrix of the ith pixel, and c is a constant.

This distance was first used in the supervised maximum likelihood (ML) classifier and showed good performances [21]. Based on this statistical measure, many

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classification methods were proposed. Some of these methods involve an important step, which is the iterative procedure.

The Wishart distance is a pseudo distance and its value could be negative in some cases. In practical applications, a well-defined dissimilarity measure is typically required to be nonnegative, identity of indiscernibles, and symmetric. The Wishart distance does not satisfy the three criteria for distance measurement.

B. RW Distance

Conradsen *et al.* [22] constructed a likelihood ratio function that is used to measure the similarity between two regions based on the Wishart distribution. The hypothesis test is

$$\begin{cases} H_0 : \mathbf{\Sigma}_i = \mathbf{\Sigma}_j \\ H_1 : \mathbf{\Sigma}_i \neq \mathbf{\Sigma}_j \end{cases} \tag{7}$$

where $\mathbf{\Sigma}_i$ and $\mathbf{\Sigma}_j$ represent the averaged coherency matrices of regions R_i and R_j, respectively. Assume that the observations are independent. Then, the likelihood ratio test statistic Q is

$$Q = \frac{L_{H0}(\hat{\mathbf{\Sigma}} \mid R_i, R_j)}{L_{H1}(\hat{\mathbf{\Sigma}}_i, \hat{\mathbf{\Sigma}}_j \mid R_i, R_j)}. \tag{8}$$

Once $\mathbf{\Sigma}_j$ is known for hypotheses H_0 and H_1, the hypothesis test becomes a general binary hypothesis test. The test statistic can be rewritten as

$$Q' = \frac{L_{H0}(\hat{\mathbf{\Sigma}}_j \mid R_i)}{L_{H1}(\hat{\mathbf{\Sigma}}_i \mid R_i)}. \tag{9}$$

By taking the negative logarithm of the likelihood ratio, the RW distance is obtained as

$$\begin{aligned} D'_{\mathrm{RW}}(i, j) &= -\frac{1}{L} \ln Q' \\ &= \ln\left(\frac{\left|\hat{\mathbf{\Sigma}}_j\right|}{\left|\hat{\mathbf{\Sigma}}_i\right|}\right) + \mathrm{Tr}\left(\hat{\mathbf{\Sigma}}_j^{-1} \hat{\mathbf{\Sigma}}_i\right) - q. \end{aligned} \tag{10}$$

If the ith region only includes one pixel, this distance can be used for superpixel segmentation and has the following form [23]:

$$D_{\mathrm{RW}}(i, j) = \ln\left(\frac{\left|\hat{\mathbf{\Sigma}}_j\right|}{|\boldsymbol{T}_i|}\right) + \mathrm{Tr}\left(\hat{\mathbf{\Sigma}}_j^{-1} \boldsymbol{T}_i\right) - q. \tag{11}$$

A few superpixel segmentation methods have used this distance, and good performance was achieved [24]. Only in the case of $\boldsymbol{T}_i = \hat{\mathbf{\Sigma}}_j$, the RW distance is equal to zero. Otherwise, the value is larger than zero. The RW distance is guaranteed to be nonnegative. However, it is not symmetric.

C. SNLL Distance

To satisfy the symmetric requirement of the distance measurement, Anfinsen *et al.* [25] proposed the symmetrized normalized log-likelihood (SNLL) distance. It is derived from the RW distance, and thus, it is usually referred to as the SRW distance, which is given by

$$\begin{aligned} D'_{\mathrm{SNLL}}(i, j) &= \frac{1}{2}\left(D'_{\mathrm{RW}}(i, j) + D'_{\mathrm{RW}}(j, i)\right) \\ &= \frac{1}{2}\left(\mathrm{Tr}\left(\hat{\mathbf{\Sigma}}_j^{-1} \hat{\mathbf{\Sigma}}_i\right) + \mathrm{Tr}\left(\hat{\mathbf{\Sigma}}_i^{-1} \hat{\mathbf{\Sigma}}_j\right)\right) - q. \end{aligned} \tag{12}$$

To measure the similarity of a pixel to a class, the SNLL distance takes the following form:

$$\begin{aligned} D_{\mathrm{SNLL}}(i, j) &= \frac{1}{2}\left(D_{\mathrm{RW}}(i, j) + D_{\mathrm{RW}}(j, i)\right) \\ &= \frac{1}{2}\left(\mathrm{Tr}\left(\hat{\mathbf{\Sigma}}_j^{-1} \boldsymbol{T}_i\right) + \mathrm{Tr}\left(\boldsymbol{T}_i^{-1} \hat{\mathbf{\Sigma}}_j\right)\right) - q. \end{aligned} \tag{13}$$

For $\boldsymbol{T}_i = \hat{\mathbf{\Sigma}}_j$, D_{SNLL} reaches its minimum value, i.e., 0. For $\boldsymbol{T}_i \neq \hat{\mathbf{\Sigma}}_j$, $D_{\mathrm{SNLL}}(i, j)$ is larger than $D_{\mathrm{RW}}(i, j)$. This distance satisfies all the three properties defined for distance measurement. In the framework of spectral clustering, the SNLL distance is useful for capturing the pairwise similarities between the pixels in the polarimetric domain [26], [27].

D. HLT Distance

The complex-kind HLT test statistic was proposed by Akbari *et al.* [20] for unsupervised change detection. The test statistic measures the similarity of two covariance matrices complying with the complex Wishart distribution. The complex-kind HLT statistic is defined as

$$\tau_{\mathrm{HLT}}(\boldsymbol{A}, \boldsymbol{B}) = \mathrm{Tr}(\boldsymbol{A}^{-1} \boldsymbol{B}) \tag{14}$$

when $\boldsymbol{A}$ and $\boldsymbol{B}$ are equal, and the value of the test statistic is equal to the polarization dimension. Changes from $\boldsymbol{A}$ to $\boldsymbol{B}$ can lead to increased or decreased values of $\tau_{\mathrm{HLT}}(\boldsymbol{A}, \boldsymbol{B})$. Thus, if the value is away from the polarization dimension, such as 3 in the case of this study, either in the larger-valued side or lower-valued side, $\boldsymbol{A}$ and $\boldsymbol{B}$ are unlikely from the same class. To eliminate the two-sided (nonsymmetric) effect when measuring the distance between $\boldsymbol{A}$ and $\boldsymbol{B}$, the reversed statistic $\tau'_{\mathrm{HLT}}(\boldsymbol{A}, \boldsymbol{B}) = \mathrm{Tr}(\boldsymbol{B}^{-1} \boldsymbol{A})$ can be combined with $\tau_{\mathrm{HLT}}(\boldsymbol{A}, \boldsymbol{B})$ to cancel the similarity direction.

Hence, to measure the distance of a pixel and a class, the symmetric HLT distance is defined as follows:

$$D_{\mathrm{HLT}}(i, j) = \max\left(\mathrm{Tr}\left(\hat{\mathbf{\Sigma}}_j^{-1} \boldsymbol{T}_i\right), \mathrm{Tr}\left(\boldsymbol{T}_i^{-1} \hat{\mathbf{\Sigma}}_j\right)\right). \tag{15}$$

As the SNLL distance, this matrix distance measure is nonnegative, indiscernible of identicals, and symmetric. However, none of the distances introduced in this section obey the subadditivity property.

III. Superpixel Segmentation

The SLIC method was proposed by Achanta *et al.* [10], which consists of two distance measures: the feature similarity defined in the CIELAB color space and the spatial proximity in the image plane space. The selection of the feature similarity measure should vary in accordance with the image statistics when SLIC is applied to different types of data. The SLIC segmentation procedure generally is implemented through three steps, i.e., initialization, iteration, and postprocessing. The initialization of clustering centers significantly affects the segmentation results. In this section, a simple and efficient initialization method is proposed. First, we first provide an overview of the general SLIC framework, in which the SLIC objective formulation is modified to adapt to the polarimetric statistical distances. Then, the reinitialization step employed in the GC-SLIC method [16] is briefly described. Finally, the new initialization method is presented.

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A. SLIC Superpixel Segmentation Framework

The idea of SLIC is to adopt the local k-means clustering technique to generate superpixels. Through localized iterative clustering, pixels with similar characteristics in adjacent space are grouped into a cluster. Each cluster is a superpixel block. The segmentation steps are summarized as follows.

1) Initialization of Cluster Centers: Suppose that N is the number of all pixels in an image and K is the number of expected superpixels. Then, the average distance between two adjacent superpixels, namely, the spatial distance between the adjacent cluster centers, should be $S = \sqrt{N/K}$ and the average size of each superpixel would be expressed as N/K. The SLIC algorithm first divides the image into K regular grids and then selects the center of each grid as the cluster center. To avoid the cluster center being chosen on image edges or on a noisy pixel, the seed location is moved to the pixel with the lowest gradient in a 3×3 neighborhood. Each cluster center corresponds to an independent label.

2) Local k-Means Clustering: This step aims to search the neighborhood of each cluster center and assign a class label to each pixel. Unlike the standard k-means that searches the whole image to update cluster centers, the SLIC confines the searching range within a $2S \times 2S$ area around the cluster center. Update cluster centers until the maximum iteration number is reached or the number of pixels switching between classes in iterations is smaller than a given number.

Set the maximum number of iterations to be itermax for instance. The labeling process of a pixel p is denoted by belong(p). The distance between a pixel p and the cluster center is denoted by dismin(p). Set the initial belong(p) as -1 and the initial dismin(p) as $+\infty$. move denotes the number of pixels changing labels in an iteration. The flowchart of the iteration procedure is given as follows.

```
while (i < itermax && move > a given number)
move = 0;
  calculate the average value for each cluster block to update
  the new cluster center;
    for u = 1 : K
      calculate the distance disnow between class u and
      pixel p in a 2S × 2S neighborhood;
        if disnow < dismin (p)
          dismin(p) = disnow;
          move = move + 1;
          belong(p) = u;
        end
    end
end
```

3) Postprocessing: In order to eliminate isolated superpixel blocks caused by noises, which usually contain only a few pixels, the postprocessing step is necessary. After local clustering, regions with fewer pixels than Nmax should be merged with the nearest superpixel block. However, to preserve strong scattering point targets, general practice is to set a threshold Nmin manually and to decide whether the merging is needed. The superpixel blocks within the size [Nmin, Nmax] are selected to calculate the dissimilarity G with their respective eight neighboring superpixel blocks. If the dissimilarity G is larger than a given threshold Gth, the block is retained; otherwise, the two blocks are merged. The dissimilarity G is defined as [17], [28]

$$G(R_i, R_j) = \frac{1}{q}\left\| \frac{\boldsymbol{T}_i^{\text{diag}} - \boldsymbol{T}_j^{\text{diag}}}{\boldsymbol{T}_i^{\text{diag}} + \boldsymbol{T}_j^{\text{diag}}} \right\| \tag{16}$$

where $\|\cdot\|$ denotes the 1-norm of a matrix, $\boldsymbol{T}^{\text{diag}}$ represents a matrix only with the diagonal elements, and q is the dimension of the vector.

4) Modified SLIC Clustering Measure: The SLIC clustering distance is a combined distance defined in a 5-D space for optical images, using the L, a, and b values of the CIELAB color space to represent the features and using the pixel coordinates to measure the spatial proximity. The feature similarity and spatial proximity are normalized individually by considering both the distribution range of the feature distance and the desired superpixel size. For optical images, Euclidean distances are applied, which are nonnegative, indiscernible of identicals, and symmetric. However, when applied to polarimetric SAR images, the feature distance in SLIC should be adapted accordingly and the combination form of the feature and spatial distances should also be modified because the statistical distances introduced in Section II are not all guaranteed to be positive. In this study, we use the following simple form to combine the two parts [29]:

$$\begin{aligned} D_{\text{SLIC}} &= \frac{d_{\text{p}}}{m} + \frac{d_{xy}}{S} \\ d_{xy}(i, j) &= \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \end{aligned} \tag{17}$$

where d_{xy} represents the spatial distance of pixels i and j and d_{p} denotes the polarimetric statistical distance. m is a tightness coefficient (or weight parameter) to adjust the ratio of the two distances. The larger m, the larger the proportion of the spatial distance, and the superpixel regions are more compact and likely to be rectangles or regular hexagons. The smaller m, the larger the proportion of the statistical distance, and the superpixel contours will be more adherent to the image edges.

B. Initialization of the GC-SLIC Method

To reduce the speckle effect on the polarimetric SAR image segmentation, Qin *et al.* [16] proposed an initialization step, which reinitializes the clustering centers based on the original regularly spaced partitioning. Details are introduced as follows. After step 1) in the SLIC framework, an extra clustering step is added to reinitialize the clustering centers. As shown in Fig. 1, suppose that $C1$–$C4$ are the original central seeds, and $R1$ with the blue boundary is the region formed by $C1$–$C4$. For each pixel p in $R1$, calculate the distance between p and C_i ($i = 1, 2, 3, 4$) and assign p to the class with the minimum distance. When all pixels in the image are relabeled, update cluster centers and continue with step 2) in the SLIC framework, i.e., the local k-means clustering.

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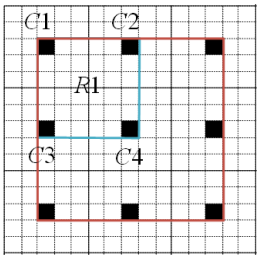

Fig. 1. Initialization step in GC-SLIC.

C. Gradient-Based Initialization

Gradient-based edge detection is a traditional vein for image segmentation, which aims to find relevant segmentation boundaries. Image gradient is useful to quantitatively evaluate the heterogeneity of image regions. The watershed approach relies on the gradient magnitudes and divides the pixels into foreground and background objects based on the topographic minima. However, the segmented blocks are often very irregular in size and shape. Hu *et al.* [30] proposed the spatial constrained watershed (SCoW) method to ensure the uniformity and compactness of the generated results by adding a spatial constraint, which is more robust and computationally efficient compared with the classic watershed.

In the SCoW method, the priority of pixel (x, y) is defined as follows:

$$\begin{aligned} p(x, y) &= p_{\mathrm{g}}(x, y) + \lambda\, p_{\mathrm{s}}(x, y) \\ p_{\mathrm{g}}(x, y) &= g_{\max} - g(x, y) \end{aligned} \tag{18}$$

where $p_{\mathrm{g}}(x, y)$ represents the gradient priority of the pixel located at (x, y) and $p_{\mathrm{s}}(x, y)$ represents the pixel priority constrained by the spatial distance. λ is a parameter balancing the two terms. $g(x, y)$ is the gradient magnitude, and $g_{\max}$ denotes the maximum possible gradient value in the image. The spatial constraint is defined as

$$\begin{aligned} p_{\mathrm{s}}(x, y) &= -d_{xy} E(x, y) \\ E(x, y) &= \exp\left(\frac{-g(x, y)}{\alpha}\right). \end{aligned} \tag{19}$$

d_{xy} denotes the Euclidean distance of a pixel (x, y) to its corresponding marker point. $E(x, y)$ is an edge protection function, and α is a damping factor. One could note that $E(x, y)$ decreases monotonically as the gradient value increases, and meanwhile, the spatial constraint priority decreases when it is close to image edges. α controls the decreasing speed of the edge protection function. The larger the value of α, the slower the decreasing speed and stronger spatial constraint.

The SCoW principle can be extended to polarimetric SAR images. In this study, we propose a new preprocessing method for the SLIC segmentation. The method is performed in a 3-D $[G, X, Y]$ space, where G is the image gradient and X and Y are the image plane coordinates. The image gradient is obtained by the constant false alarm rate (CFAR) edge detector [31]. The priority definition is similar to that of SCoW in form but has adaptive changes to be suitable for polarimetric SAR images. The distance between pixel p and class center u is defined as

$$D_{\mathrm{g}}(u, p) = d_{\mathrm{polg}}(u, p) + \lambda\, d_{\mathrm{pols}}(u, p) \tag{20}$$

where the gradient distance $d_{\mathrm{polg}}(u, p)$ is given by

$$d_{\mathrm{polg}}(u, p) = |\, g_{\mathrm{CFAR}}(u) - g_{\mathrm{CFAR}}(p)\,|. \tag{21}$$

$g_{\mathrm{CFAR}}(p)$ refers to the gradient of p, i.e., the edge possibility calculated by the CFAR edge detector.

The spatial distance $d_{\mathrm{pols}}(u, p)$ is defined as

$$\begin{aligned} d_{\mathrm{pols}}(u, p) &= d_{xy}(u, p) E_{\mathrm{pol}}(p) \\ E_{\mathrm{pol}}(p) &= \exp\left(\frac{g_{\mathrm{CFAR}}(p)}{\alpha}\right). \end{aligned} \tag{22}$$

$d_{xy}(u, p)$ is the spatial Euclidean distance between u and p in the image plane. $E_{\mathrm{pol}}(p)$ is the edge protection function and α controls its decreasing velocity.

Basically, the gradients of pixels located at image edges are larger than those of pixels in smooth regions. The edge protection function is exponentially increased. The larger the gradient value of p, the larger $E_{\mathrm{pol}}(p)$ it is. A larger $d_{xy}(u, p)$ indicates p is less likely to be classified into class u. The growth rate of $E_{\mathrm{pol}}(p)$ is controlled by an appropriate order of magnitude α. In (22), α indirectly serves as an adaptive weight parameter for the spatial Euclidean distance. By amplifying the distance between the edge area and the central seed, pixel points at the edges are preserved and pixel points are clustered within homogeneous areas.

The SCoW method was implemented via a marker-controlled segmentation [32]. However, due to the effects of noises and region discontinuities, marker-controlled watershedding does not exhibit good boundary adherence. In this study, we propose a simple segmentation method. The procedure is schematically given in Fig. 2. First, select cluster centers at regular intervals. This step is the same as the SLIC initialization. The black grids in Fig. 2(a) are initial cluster centers. Second, assign S^2 pixels to each cluster center. Taking cluster center $C1$ as an example, select S^2 pixel points in the dark blue box, which should be near in spatial proximity and similar in the feature space to $C1$. Then, label these pixels as $C1$, such as the blue girds in Fig. 2(b) and the green grids in Fig. 2(c). After traversing all the cluster centers, some pixels may be labeled more than once. For example, the gray grids in Fig. 2(d) are marked by both $C1$ and $C2$. These pixels result in overlapping regions and need to be divided such that each pixel has a single label. Assume that p' is a pixel in the overlapping region. Assign p' to the closest cluster center. After all pixels in the overlapping region are reassigned, the initialization is complete.

Details are summarized as follows.

1) Set values for K, λ, and α.
2) Initialize cluster centers $U_i = [G_i, x_i, y_i]^{\mathrm{T}}$ by sampling pixels on regular grids with interval S.
3) Move cluster centers to the lowest gradient position in an $n \times n$ neighborhood.
4) For each cluster center u, calculate the distances $D_{\mathrm{g}}(u, p)$ for pixels in the $2S \times 2S$ neighborhood

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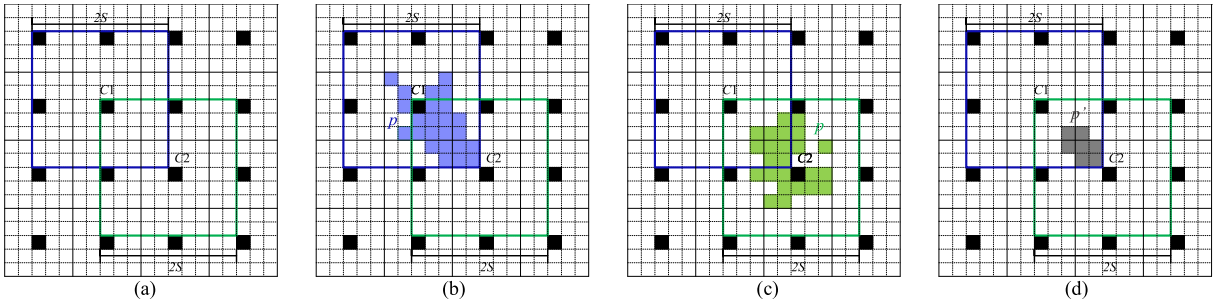

Fig. 2. Proposed initialization steps for SLIC. (a) Regular grids. (b) Re-classification using (20) for center $C1$. (c) Re-classification using (20) for center $C2$. (d) Pixels labeled by both $C1$ and $C2$.

using (20). Then, a distance set is obtained as $D_g(u) = [D_g(u, p_1), D_g(u, p_2), \ldots, D_g(u, p_{2S\times 2S})]$.

5) Sort $D_g(u)$ in the ascending order. Select the first S^2 pixels and label them as u.
6) For pixel p' that has been labeled more than once, calculate the distance $D_g(u_i, p')$ using (20). Note that the cluster center $u_i (i = 1, 2, \ldots)$ resides in p''s $2S \times 2S$ neighborhood. Then, it is obtained that $D_g(p') = [D_g(u_1, p'), D_g(u_2, p'), \ldots, D_g(u_c, p')]$, where c is the relabeled times for p'.
7) Assign pixel p' to the cluster with minimum value in $D_g(p')$.

IV. Experiments

To demonstrate the effectiveness of the proposed method, extensive experiments are conducted on polarimetric SAR data sets collected by AIRSAR and RADARSAT-2. Validations are carried out via three groups of experiments to successively evaluate the following aspects: 1) contribution of the initialization step; 2) performances of the four statistical measures; and 3) performances of four SLIC-based algorithms (i.e., SLIC [29], GC-SLIC [16], IER [17], and the proposed method), where the Ncut [26] and TurboPixel [33] algorithms are also compared. In Section IV-A, the test data sets and the commonly used evaluation indexes are first introduced.

A. Data Sets and Evaluation Parameters

AIRSAR data sets acquired over Death Valley, CA, USA, in September 1989, and over an agriculture field in Flevoland, The Netherlands on July 3, 1991, are used for demonstration, which have 1279×1024 pixels and 700×500 pixels, respectively. The third data set is from RADARSAT-2, acquired over an agriculture field in Wallerfing, Germany, on May 28, 2014, which has 500×500 pixels. Pauli color-coded images of the test data sets are shown in Fig. 3. For visual inspection and quantitative evaluation, manual segmentation maps used as ground truth are provided for the agriculture field data sets. The Wallerfing data has real ground-truth values for the agricultural land, so we only depicted the boundaries for the river area. All data sets are filtered by the Lee filter with window size 9 before implementing the experiments.

To evaluate the performances of different methods, three indexes are used to evaluate the segmentation results, which are boundary recall (BR) rate, undersegmentation error (USE), and running time (RT). The BR and USE are explained as follows.

BR is the ratio of boundary pixels shared by both the obtained superpixels and the ground truth, which is defined as [34]

$$\mathrm{BR} = \frac{N_{\mathrm{S}\cap\mathrm{G}}}{N_{\mathrm{G}}} \tag{23}$$

where $N_{\mathrm{S}\cap\mathrm{G}}$ denotes the number of pixels on both the superpixel edges and the ground-truth edges and N_{G} is the number of pixels of the ground-truth edges. A larger BR indicates that superpixel blocks agree better with the image edges.

According to the superpixel evaluation criteria, one superpixel block should only occupy one object. The deducting area is defined as the superpixels that occupy more than one ground-truth segments. The USE is computed by averaging the deducting areas over the whole image, which is defined by [10]

$$\mathrm{USE} = \frac{1}{N}\left[\sum_{i=1}^{M}\left(\sum_{[s_j|s_j\cap g_i > B]}|s_j|\right) - N\right] \tag{24}$$

where N is the number of pixels in an image, g_i is a ground-truth segment, and s_j is an obtained superpixel. $s_j \cap g_i$ is the number of pixels in the overlapping area of g_i and s_j. B is a minimum number of pixels in s_j overlapping g_i. $|\cdot|$ denotes the number of pixels in a superpixel. A lower value of USE indicates that the segmentation result is more accurate.

B. Performance of Initializations

In this section, we compare the SLIC method, the GC-SLIC method, and the proposed method. SLIC has no initialization step, whereas GC-SLIC and our method have initialization steps. For consistent analysis and fair comparisons, we choose the Wishart distance with the modified clustering function for the three algorithms. Since the number of superpixels has an influence on the final segmentation result, in practice, the number of superpixels K is usually set according to the complexity of the image. In order to make a comprehensive comparison, a set values of K, i.e., $K = \{100, 200, 300, 400, 500, 1000, 1500, 2000, 2500, 3000, 3500, 4000, 4500, 5000\}$, are applied. The tightness coefficient m is set to 2 and α is set to 100.

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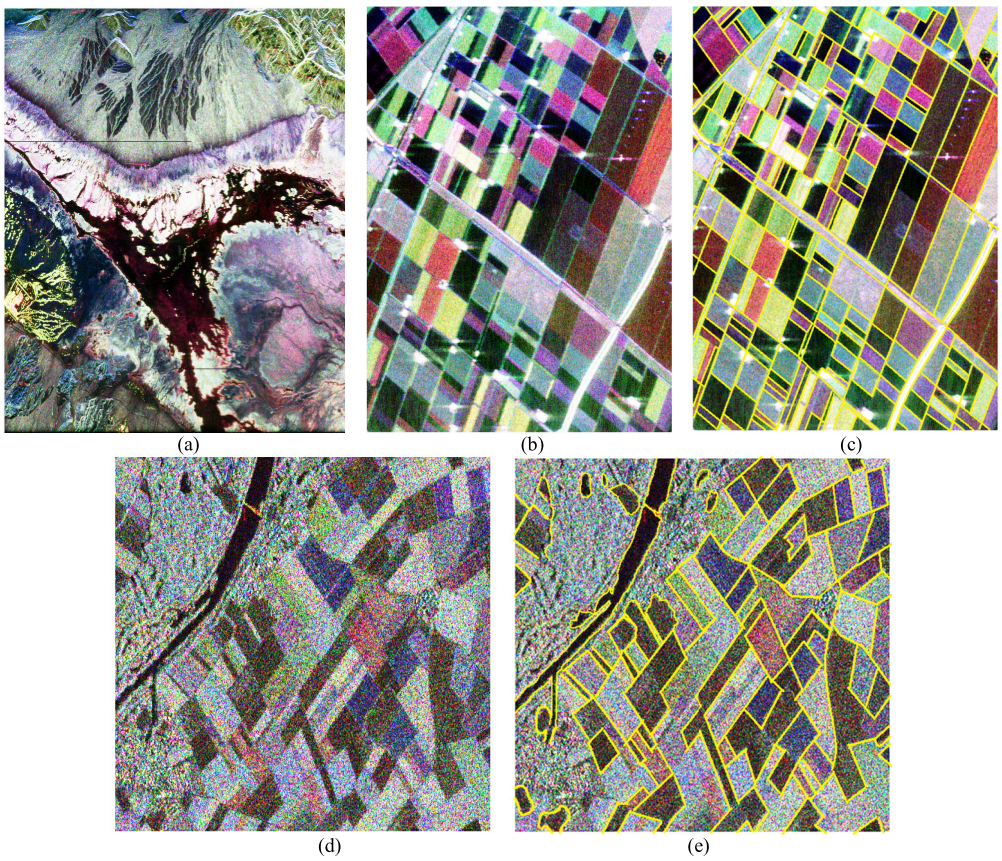

Fig. 3. Pauli color-coded images of the test data sets. (a) Death Valley data. (b) Flevoland data. (c) Ground-truth map of the Flevoland data. (d) Wallerfing data (May 28, 2014). (e) Ground-truth map of the Wallerfing data.

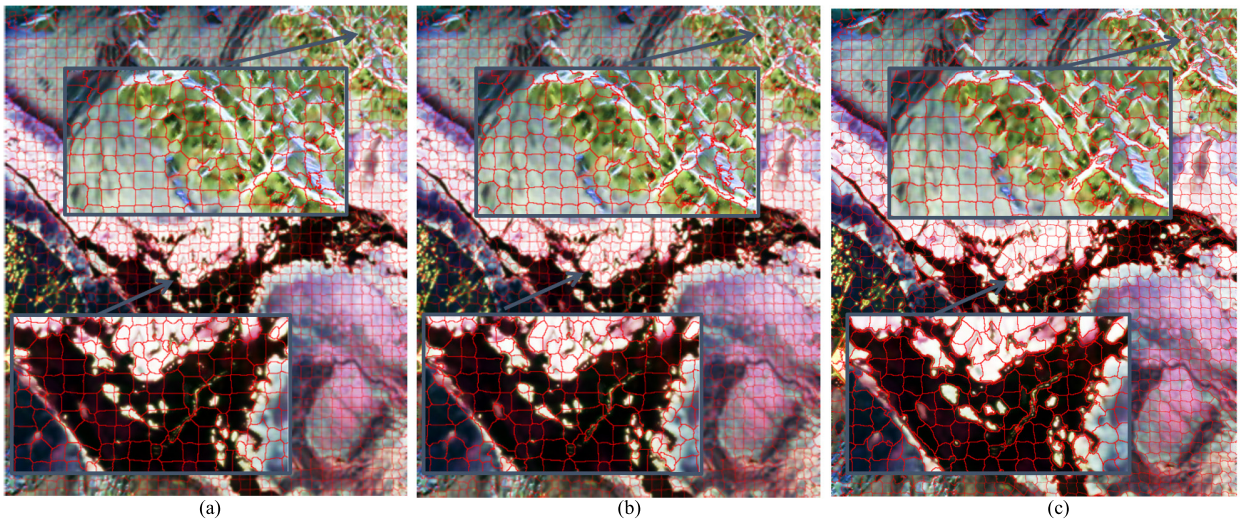

Fig. 4. Superpixel segmentation results for the Death Valley data. (a) SLIC. (b) GC-SLIC. (c) Proposed method.

1) Death Valley Data: Fig. 4 shows the segmentation results. The number of superpixels is 2000. Superpixels in Fig. 4(a)–(c) are generally similar in shape and size. We zoomed-in view two areas with complex landscapes for detailed comparison, shown in the gray rectangle boxes. In Fig. 4(a) and (b), many superpixels contain a mixture of different objects, as evidenced by the nonuniform colors presented within a single superpixel. The mountain ridge areas are not accurately located, as is shown that superpixel boundaries are not aligned with the ridge edges. Comparatively in Fig. 4(c), superpixels generated by the proposed method coincide with the pattern edges. Image details, such as point targets, areas with linear features, and irregularly shaped small areas, are much better preserved. Results show that

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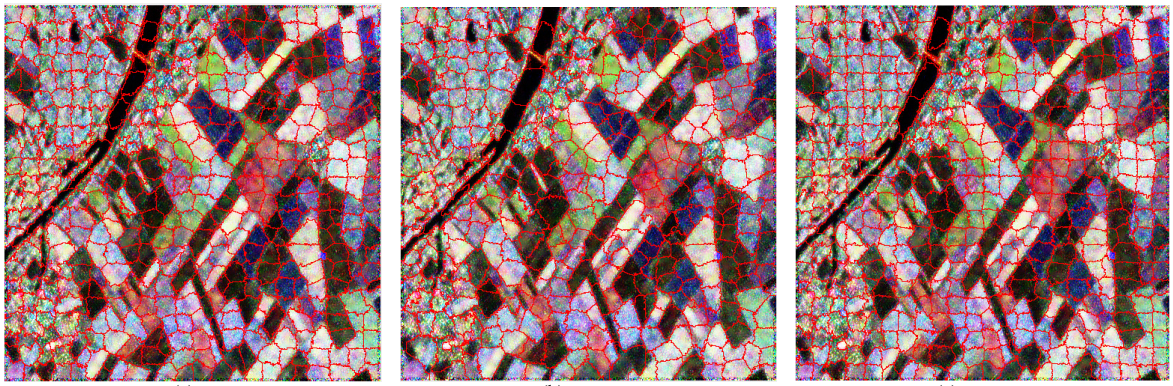
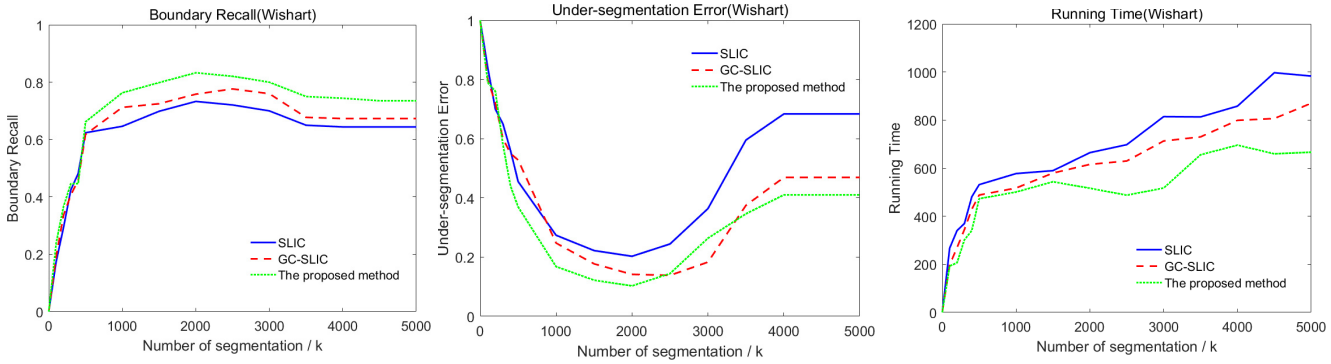


(a) (b) (c)

Fig. 5. Superpixel segmentation results for the Wallerfing data (May 28, 2014). (a) SLIC. (b) GC-SLIC. (c) Proposed method.



(a) (b) (c)

Fig. 6. Performance comparison (the Wallerfing data, May 28, 2014). (a) BR. (b) USE. (c) RT.

segmentation is greatly affected by the initial seeds. In the SLIC method, some initial class centers may be located at the edges of the regions, and thus, in the subsequent iterative clustering, the class centers might oscillate between the neighboring segments, which will result in the results lack of adaptiveness to region complexity. The performance of GC-SLIC is better than that of SLIC, but still superpixel boundaries are not adhered closely to small areas with irregular shapes. Some image details such as the linear features are not preserved. With the proposed initialization method, local clustering produces promising segmentation results.

2) Wallerfing Data: Fig. 5 shows the superpixel results for the Wallerfing data with 500 superpixels. The corresponding BR rate, USE, and RT indexes are shown in Fig. 6. With the increasing segmentation number K, the BR of the three methods is greatly improved at the beginning until an appropriate number is reached. The reason is that as the number of superpixels increases, superpixel blocks become smaller and will be more adherent to image boundaries. The knee identified in the curve for this data is around 1000. When K is larger than 2000, there is little improvement on BR. In Fig. 6(a), it shows that the BR of the proposed method is always higher than the other methods. The BR value of the proposed method can reach 0.83 when K is 2000, which is about 0.08 higher than that of GC-SLIC and 0.06 higher than that of SLIC. The curve of USE first decreases and then increases with the increasing K. When K is within the range [1000, 3000], USE is with relatively small values. In Fig. 6(b), the lowest USE value of the proposed method can reach 0.1, while the lowest USE value of GC-SLIC and SLIC can reach 0.14 and 0.2, respectively. Comparatively, the proposed method has a small USE. It is observed from Fig. 6(c) that RT increases with K, and the proposed method has a lower RT for clustering than the SLIC and GC-SLIC methods since stable grouping seeds can reduce the iteration number.

C. Performance of Distance Measures

To explore the effect of the distance measurement on segmentation results, the statistical distances introduced in Section II are compared and evaluated. Except for the HLT distance, the other distances have already been applied to polarimetric SAR image segmentation, in which only the Wishart and RW distances have been studied for SLIC superpixel. The four distances are embedded in the proposed method since it obtained the best results in Section IV-B. For the Wallerfing data, the tightness coefficient of the RW distance is set to 0.3 and the tightness coefficients of the Wishart, SNLL, and HLT distances are set to 2. By observing the curves in Fig. 6, we found that when K is less than 500 or larger than 3000, the segmentation result is not good. Therefore, K is set from 500 to 3000 for the Wallerfing data. For the Flevoland data, the tightness coefficients of the Wishart and the RW distances are set to 0.2, the tightness coefficients of

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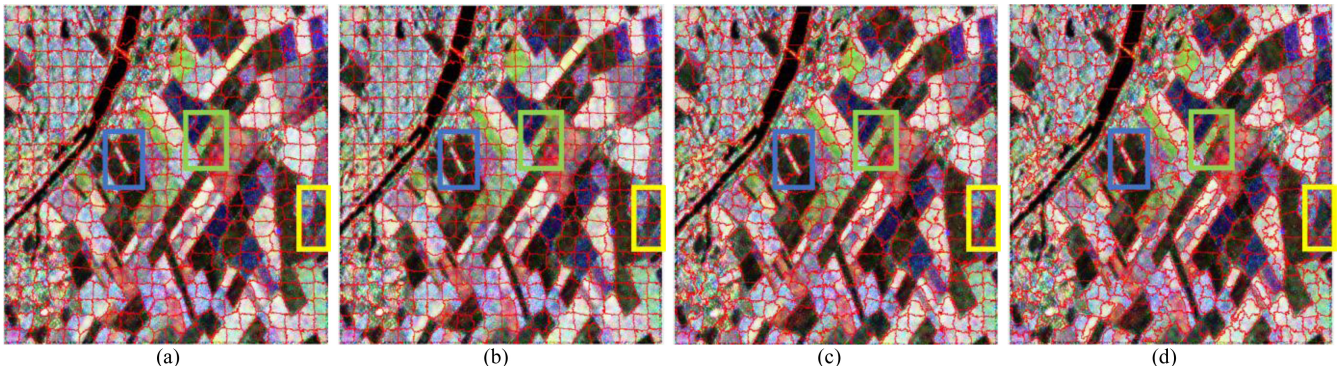

Fig. 7. Superpixel segmentation results for the Wallerfing data (May 28, 2014). (a) Wishart. (b) RW. (c) SNLL. (d) HLT.

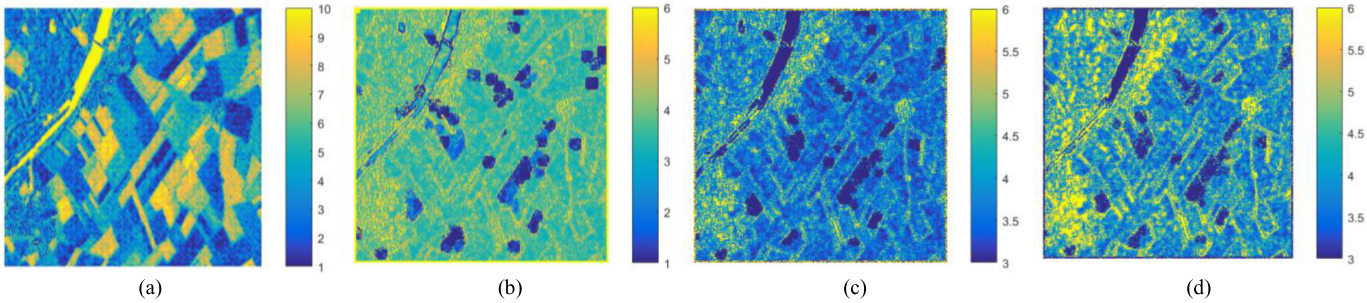

Fig. 8. Final distance between a pixel and its clustering center (the Wallerfing data, May 28, 2014). (a) Wishart. (b) RW. (c) SNLL. (d) HLT.

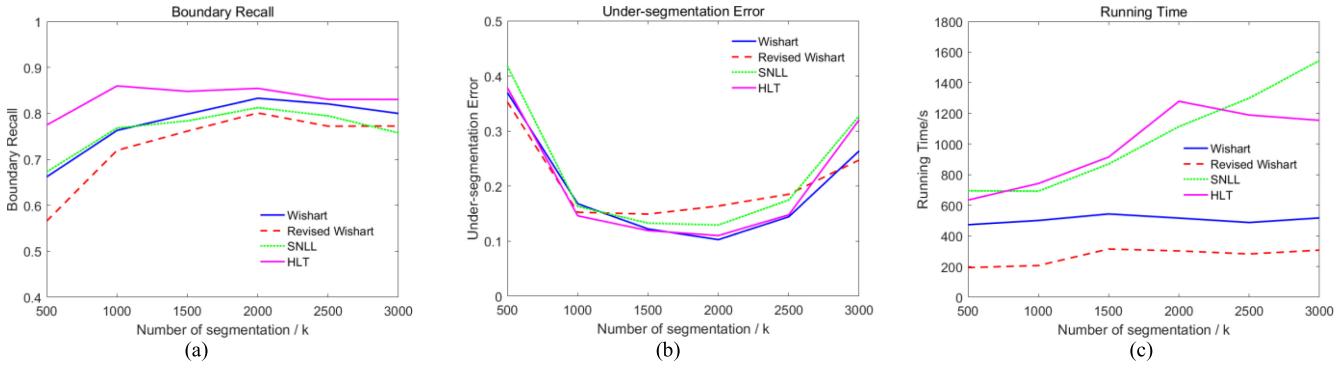

Fig. 9. Performance comparison (the Wallerfing data, May 28, 2014). (a) BR. (b) USE. (c) RT.

the SNLL and HLT distances are set to 0.7, and K is set from 500 to 5000. α remains the same as 100.

1) Wallerfing Data: Fig. 7 shows the segmentation results of the Wallerfing data with 500 superpixels. The superpixels in Fig. 7(a) and (b) are similar in both size and shape. In Fig. 7(c) and (d), the superpixel blocks in homogeneous regions are obviously larger than those in Fig. 7(a) and (b). We outlined some details for illustration. It is observed that the narrow strip of farmland in the blue rectangle is not correctly segmented in Fig. 7(a) and (b), and the fields' boundaries are not accurately located as shown in the green and yellow rectangles. In comparison, results in Fig. 7(c) and (d) show better segmentation boundaries, preserving small strip features.

Fig. 8 shows the final statistical distance of each pixel to its class center, denoted by F-D in this article. Fig. 9 shows the quantitative evaluation for the four distances. The F-D image of the Wishart distance mostly agrees with our expectation. Each block has neat edges and uniform colors, indicating that pixels in the same ground object have the same distance to the class center. In addition, different ground objects have different colors and thus can be easily distinguished from the ambient areas. Therefore, in Fig. 9(b), the USE of the Wishart distance has the lowest value of 0.10. In Fig. 8(b)–(d), boundaries of the ground objects have relatively larger values in the F-D images than the flat areas, indicating that for the RW, SNLL, and HLT distances, the edge information takes major effect in the iterative segmentation. However, in Fig. 8(b), it shows that for a majority part of the image, the RW distance values do not vary too much, distributed in a limited range, and edges are not obviously observed, which is evidence for the poor performances of RW with the lowest BR value and the highest USE value as indicated in Fig. 9(a) and (b). In the SNLL and HLT F-D images, edges are obviously clearer, and edges are presented better in the HLT distance map. In Fig. 9, the HLT

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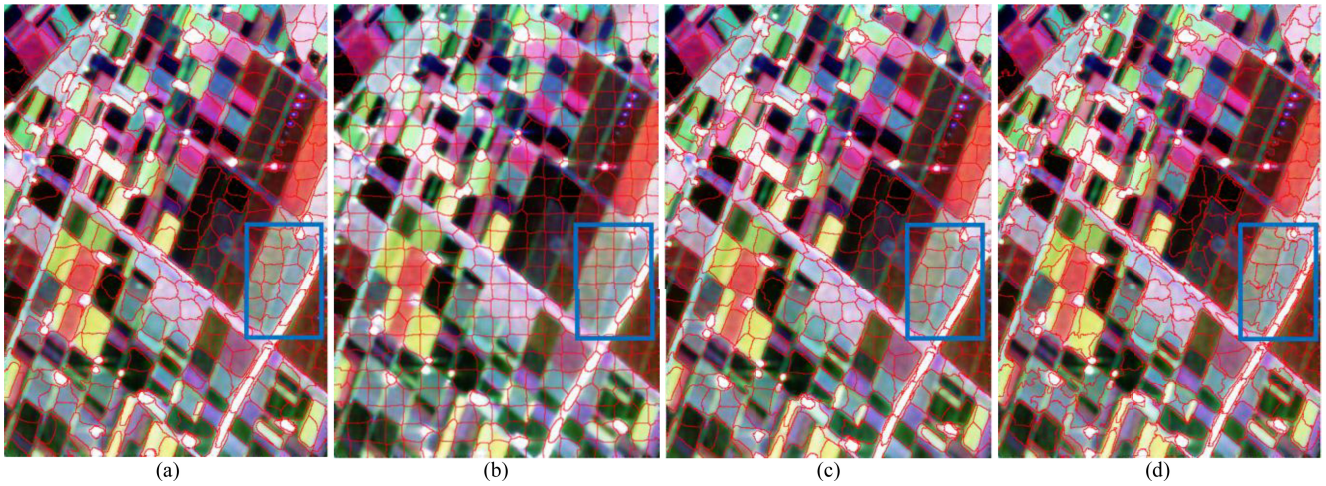

Fig. 10. Superpixel segmentation results for the Flevoland data. (a) Wishart. (b) RW. (c) SNLL. (d) HLT.

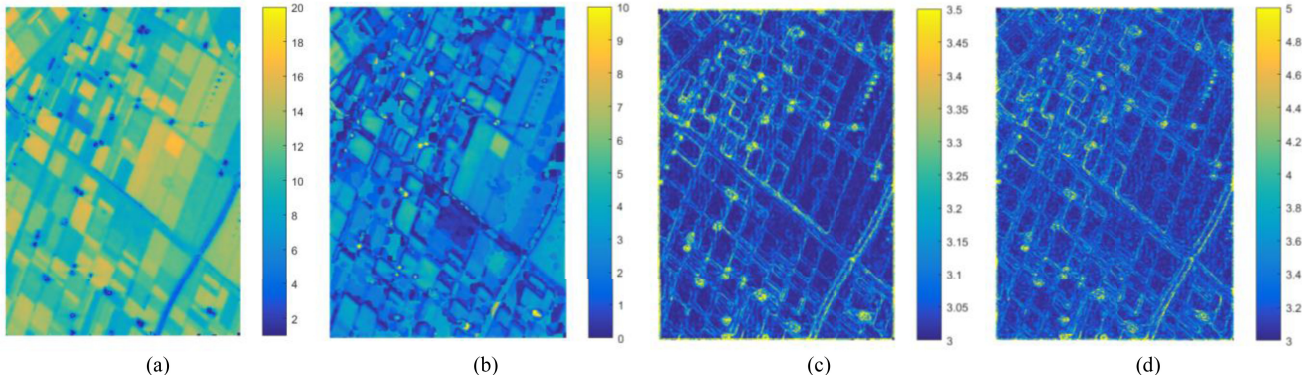

Fig. 11. Final distance between a pixel and its clustering center (the Flevoland data). (a) Wishart. (b) RW. (c) SNLL. (d) HLT.

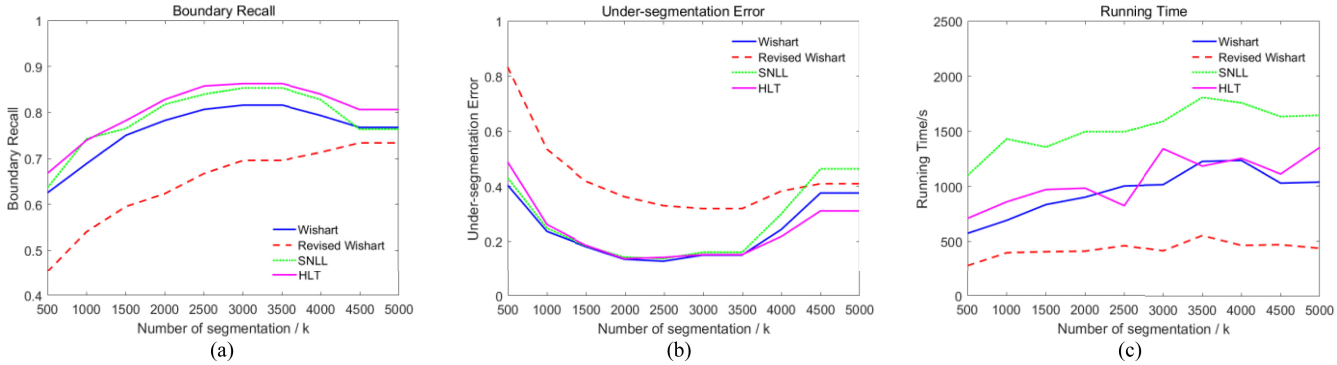

Fig. 12. Performance comparison (the Flevoland data). (a) BR. (b) USE. (c) RT.

distance has the highest BR rate, and the USE index is quite similar to that of the Wishart distance. Both HLT and SNLL need more convergence times, but the maximum BR value and the minimum USE value of the SNLL are 0.04 lower and 0.05 higher than those of the HLT, respectively. Generally, the HLT distance produces the best segmentation results, which is slightly better than the Wishart distance. However, the Wishart distance is more computationally efficient.

2) Flevoland Data: Fig. 10 shows the segmentation results for the Flevoland data, and Fig. 11 shows the F-D maps. It is seen that the result in Fig. 10(b) is the worst, with only a few superpixel contours aligned with the image boundaries. The RW distance does not exhibit effective performance for this data set. Superpixels in Fig. 10(a), (c), and (d) are better in the visual inspection. Superpixels generated by the three distances adhere closely to the boundaries of the fields. The outlined area has two pieces of fields. It shows that only the HLT distance can correctly locate the field boundaries, which is also an evidence for the highest BR rate obtained by HLT in Fig. 12(a).

In Fig. 11(a), the color of each area is uniform, that is, pixels belonging to the same class present similar statistical characteristics by the Wishart distance. However, the boundaries between different fields are undistinguishable in some regions.

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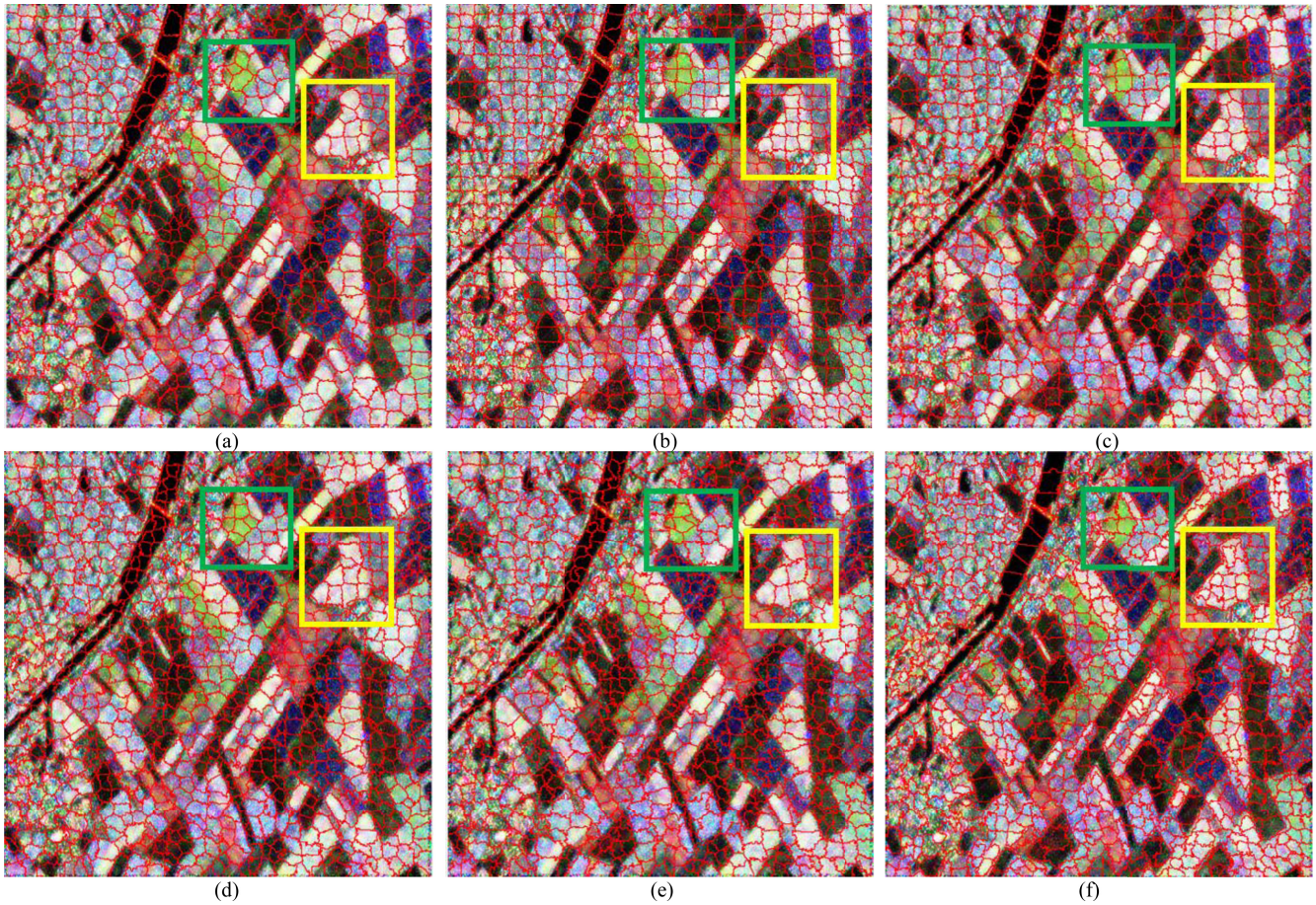

Fig. 13. Superpixel segmentation results for the Wallerfing data (May 28, 2014). (a) GC-SLIC, Wishart. (b) IER, Wishart. (c) Proposed method, Wishart. (d) GC-SLIC, HLT. (e) IER, HLT. (f) Proposed method, HLT.

This can be explained by the poor BR rate in comparison with the HLT distance in Fig. 12(a) and good USE obtained in Fig. 12(b). In Fig. 11(c) and (d), edges are better preserved, which indicates that HLT and SNLL will have better BR rates. SNLL and HLT have the maximum BR values of 0.86 and 0.87, respectively, and their minimum USE values are 0.13. For the result with the Wishart distance, maximum BR and minimum USE values are 0.81 and 0.12, respectively. For this data set, the HLT and Wishart distances have similar RTs. Although the result of HLT has a higher BR rate, superpixels generated by the Wishart distance are more compact and uniform.

D. Performance of Superpixel Algorithms

Based on the SLIC framework, there are several improved methods. Besides the GC-SLIC that has already been analyzed in Section IV-B, the recently developed IER algorithm is also included for comparison. GC-SLIC and the proposed method adopt different initialization steps. IER has no initialization processing, and its improvement on SLIC is by selecting unstable points during the iterative clustering to accelerate the convergence speed. Note that in the original GC-SLIC [16] and IER methods [17] for polarimetric SAR imagery, the RW distance is used. Note that in Section IV-B, the SLIC and GC-SLIC algorithms are slightly modified by using the proposed clustering function in (17). It is found that with the modified clustering function, the Wishart distance performs better than the RW distance in both SLIC and GC-SLIC algorithms. Based on the RW distance, we also compared the performances of the proposed method with the original GC-SLIC and IER methods. Results indicated that the proposed method is better than the GC-SLIC and IER methods. In fact, if the positive-guaranteed RW distance is used, performances will not be affected by the squared or nonsquared measures. Hence, considering the superior performances of the Wishart and HLT distances obtained in Section IV-C, only these two distances are embedded in the SLIC-based algorithms for comparison in this section. All parameter settings are the same as those in Section IV-C.

Fig. 13 shows the segmentation results of the Wallerfing data. For the three algorithms, it is observed that superpixels in Fig. 13(a)–(c) generated by the Wishart distance are visually more compact and relatively regular in shape and size in comparison with those in Fig. 13(d)–(f) by the HLT distance. The GC-SLIC and IER algorithms produce better results with the Wishart distance. Two areas are outlined for detailed comparison. In the yellow rectangle, the field boundaries are delineated better by the proposed method than those by the other two methods. In the green rectangle, there exist small

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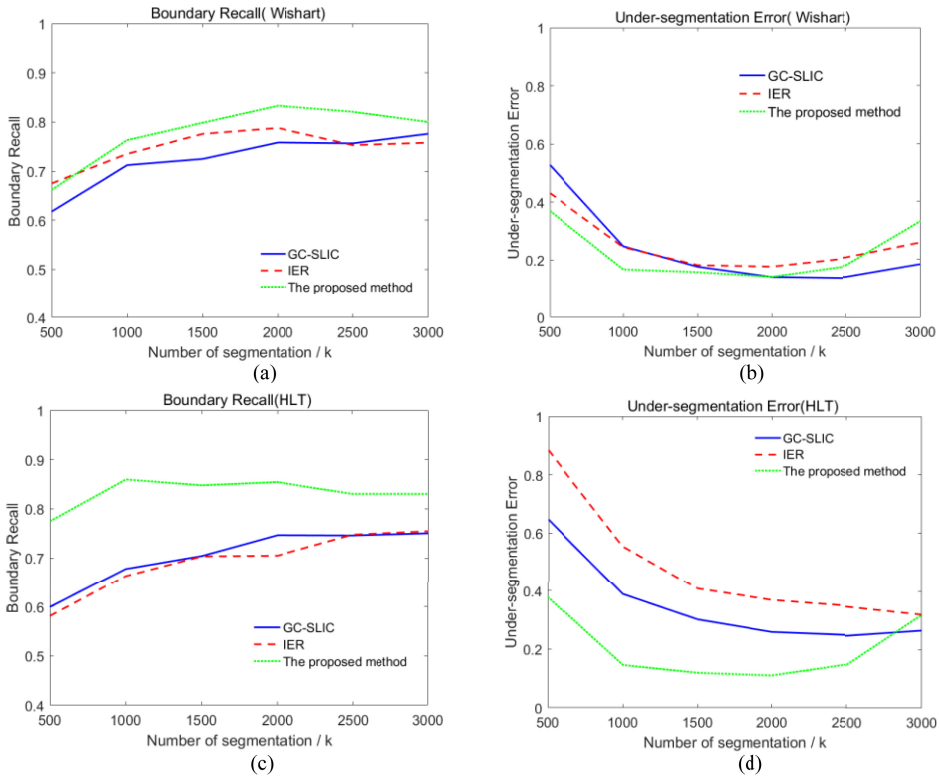

Fig. 14. Performance comparison (the Wallerfing data, May 28, 2014). (a) BR (Wishart). (b) USE (Wishart). (c) BR (HLT). (d) USE (HLT).

strips of farmland and some field edges are weak. For this area, superpixels generated in Fig. 13(f) fit the image boundaries best and are adaptive to the field sizes, even for those with irregular shapes.

Quantitative evaluation is given in Fig. 14. It shows again that for both Wishart and HLT distances, an improvement on the initial centers can significantly improve the results. The proposed algorithm always has higher BR rates, especially when it is combined with the HLT distance. In Fig. 14(b), it shows that when the superpixel number is larger than 2000, the GC-SLIC has a lower USE value. However, when the superpixel size is too small, the USE index will be of little significance. By comparing Fig. 14(a) and (c), it shows that the HLT can achieve good overall boundary adherence. However, from Fig. 13(c) and (f), it is also observed that superpixels generated by the HLT have nonuniform sizes and shapes, which vary with the ground object size, alignment, and homogeneity. Comparatively, the Wishart distance can generate relatively uniform superpixels but is slightly inferior in image edge preservation.

Besides the test data sets shown in Fig. 3, validation is also performed on another four data sets, including the multiphase data acquired on June 21, 2014, July 15, 2014, and August 18, 2014, over the same area as shown in Fig. 3(d) by RADARSAT-2, and the data from AIRSAR over San Francisco. The San Francisco image has 900×1024 pixels. TurboPixel [33] and Ncut [26] methods are also included for comparison. Ncut is a graph partitioning-based algorithm. The objective is to find the minimum similarity of candidate partitions, which is defined in the matrix distance space. The pixel-based Ncut algorithm has high time complexity. Thus, we adopted the block processing scheme as that used in [26] for practical implementation. TurboPixel is a geometric analysis-based algorithm, which incorporates the geodesic active contour model and is implemented via level set methods. The local affinity measure is defined based on the grayscale intensity gradient. In other words, the boundary evolution is driven by the image gradient. In the results shown in Fig. 15, the proposed method and the Ncut method are carried out on the coherency matrix and the TurboPixel algorithm is performed on the diagonal terms of the coherency matrix. In this experiment, the Wishart distance is used for SLIC. The superpixel numbers for the Death Valley and San Francisco images are 1000, Wallerfing images around 200. It is clearly observed that TurboPixel is not suitable for images with complex landscapes. In the results of the Death Valley data, TurboPixel could not capture the linear structures, locate mountain ridges, and accurately indicate the boundaries of heterogeneous areas. For the farmland scene and the San Francisco area, TurboPixel tends to ignore small, linearly shaped objects. Results of the Ncut algorithm are inferior to those of the TurboPixel and SLIC algorithms, especially for heterogeneous areas, such as the forested areas in the San Francisco image. Ncut could not accurately segment the low backscattered small areas in forests. At least for the test data sets of this study, the Ncut algorithm does not perform well. Furthermore, since the block processing scheme is adopted, there may have the superpixel boundary malposition. By visual inspection, the results of the proposed SLIC method is superior in boundary adherence, generating uniform and compact

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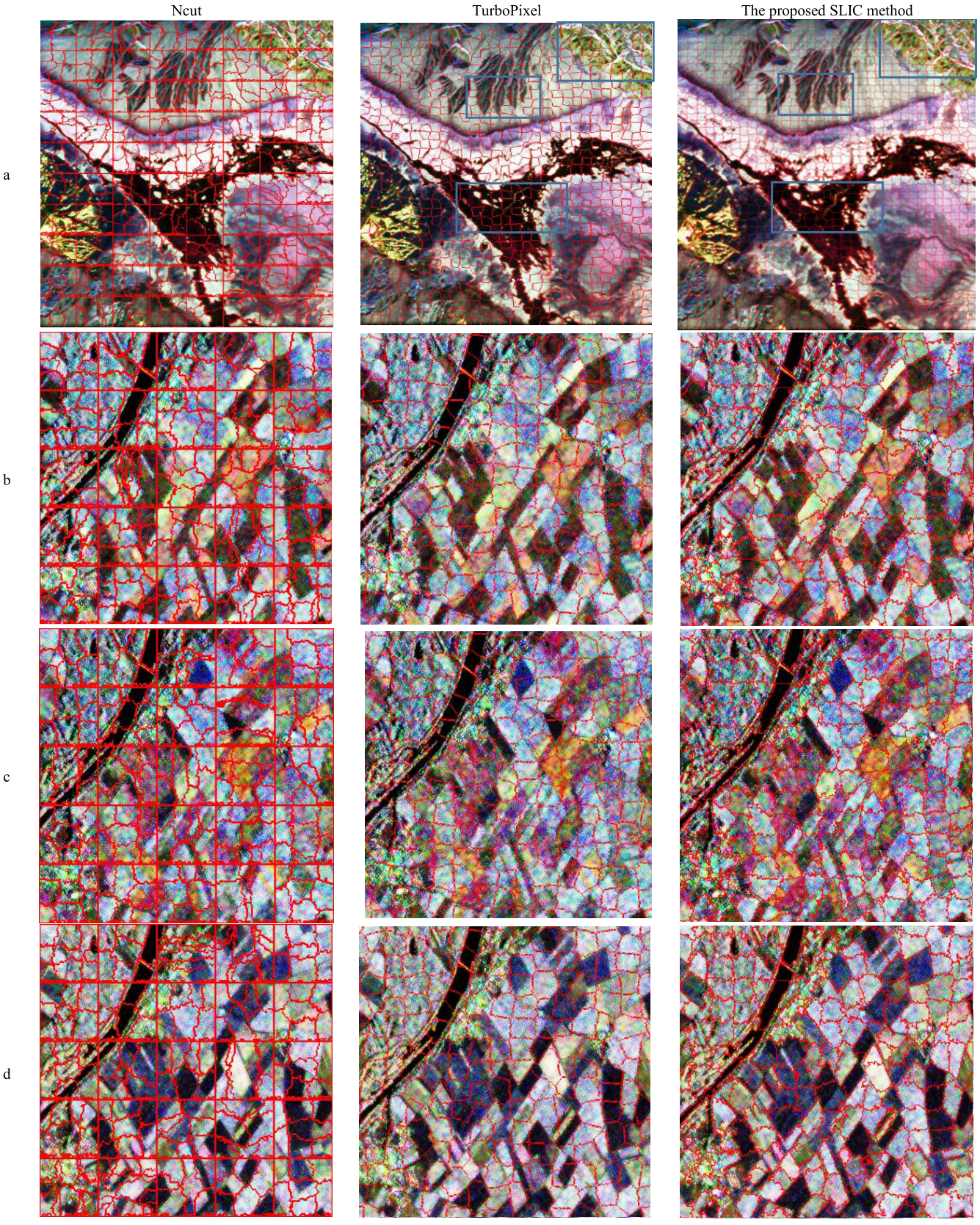

Fig. 15. Superpixel segmentation results. (a) Death Valley data. (b) Wallerfing data (June 21, 2014). (c) Wallerfing data (July 15, 2014). (d) Wallerfing data (August 18, 2014). (e) San Francisco data (AIRSAR, 1988).

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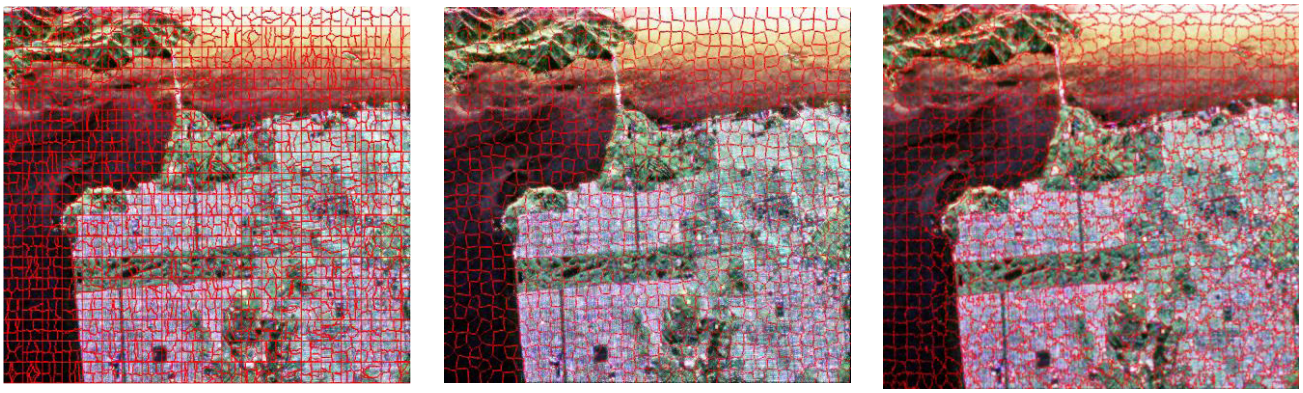

Fig. 15. *(Continued.)* Superpixel segmentation results. (a) Death Valley data. (b) Wallerfing data (June 21, 2014). (c) Wallerfing data (July 15, 2014). (d) Wallerfing data (August 18, 2014). (e) San Francisco data (AIRSAR, 1988).

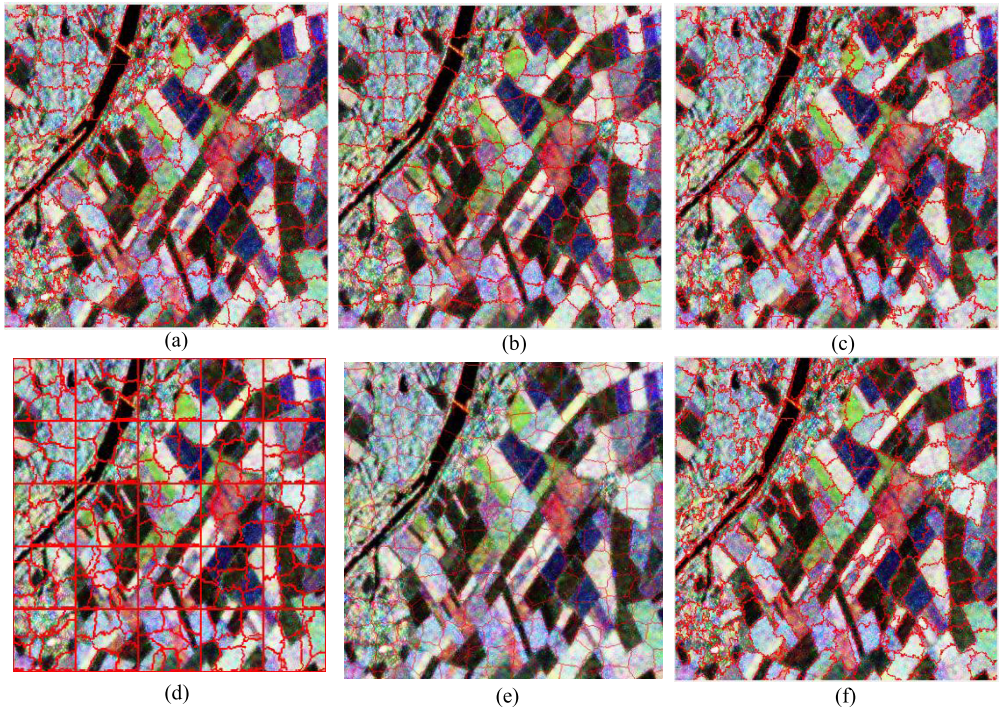

Fig. 16. Superpixel segmentation results for the Wallerfing data (May 28, 2014). The Wishart distance embedded in the modified clustering function is used for the SLIC-based methods. (a) SLIC. (b) GC-SLIC. (c) IER. (d) NCut. (e) TurboPixel. (f) Proposed SLIC method.

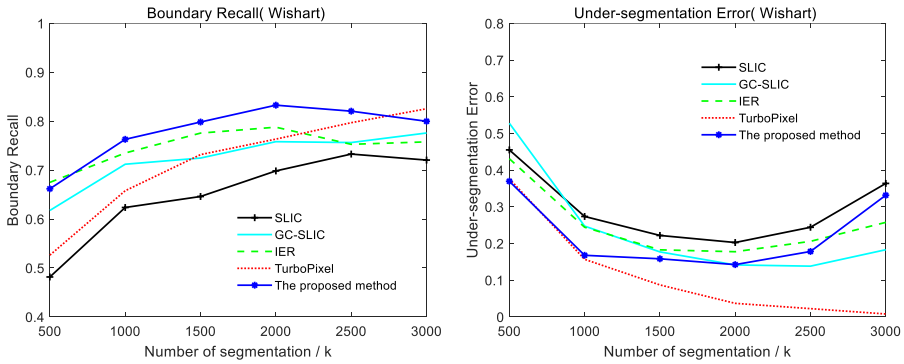

Fig. 17. Performance comparison (the Wallerfing data, May 28, 2014).

superpixels. Various test sites also show the robustness and effectiveness of the proposed method for different surveillance scenarios.

Finally, we compare the performances of the SLIC, Ncut, and TurboPixel methods. The segmentation results are given in Fig. 16, and quantitative evaluation is given in Table I and Fig. 17. For the SLIC, GC-SLIC, IER, and the proposed methods, the Wishart distance embedded in the modified clustering function is used for calculating superpixels. From Table I, it is observed that the SLIC-based algorithm has

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TABLE I
BR AND USE (THE WALLERFING DATA, MAY 28, 2014)

K=200	SLIC[29]	GC-SLIC[16]	IER[17]	NCut[26]	TurboPixel[33]	The proposed method
BR	0.6034	0.4704	0.5445	0.2934	0.2950	**0.6289**
USE	0.1617	0.2985	0.1670	0.5214	0.1758	**0.1594**

better segmentation results than both Ncut and TurboPixel, and the proposed SLIC method performs best. Fig. 17 shows the variations of USE and BR indices with the superpixel numbers. Note that the Ncut method is not included since the computational cost is heavy and it does not perform well on these data sets. It shows that the proposed method has the best boundary adherence. With respect to the USE index, when the superpixel number is really large, this index is of less significance. The BR value is a more trustworthy evaluation index.

V. CONCLUSION

The SLIC superpixel approach was studied for polarimetric SAR images. Both the segmentation procedure and the clustering measure were studied carefully with the aim to provide superior and robust results. To provide stable center seeds, a new initialization method, which aims to find pixels with closest gradients in homogeneous neighborhoods, was proposed based on the spatial-constraint watershed algorithm. Results demonstrated on superpixel boundary adherence. Also, point targets, small objects with irregular shapes, linear and curved features are better preserved comparing to those using previous initialization methods. In addition, various test sets, including data from agriculture fields, mountainous areas, and urban areas, demonstrated the segmentation robustness and effectiveness of the proposed method. The clustering function is also modified, which cancel the necessity of the nonnegativity of the feature similarity, to adapt to various polarimetric statistical distances. By this means, the commonly used polarimetric test statistics can be unified in the same framework for a fair comparison.

Four classic statistical distances, including the nonpositive guaranteed Wishart distance, nonnegative RW distance, SRW distance, and the HLT distance, are embedded in the existing segmentation flowcharts, such as the SLIC, IER, and GC-SLIC methods. Note that since the original clustering measures of IER and GC-SLIC could not directly incorporate the Wishart distance, we use the modified clustering function of this article to substitute their clustering measures. This will not only extend the adaptiveness of both IER and GC-SLIC to incorporate more statistical similarities but also not affect the segmentation performances. By comparing the statistical distances in the same procedures, we found that for all the methods, including SLIC, GC-SLIC, IER, and the proposed method, superpixels generated using the Wishart distance are more compact and regular in shape and size and those using the HLT distance adhere more precisely to image boundaries than the other distances. Performances of the RW distance are worst among the four test statistics. To our knowledge, the HLT distance is applied for the first time to image segmentation. With the proposed method, the HLT outperforms other distances in performance improvement.

This study shows that the initial cluster centers are of importance to improve the segmentation performance of the SLIC. The local clustering measure is modified to adapt to the characteristics of polarimetric SAR data, which is simple and suitable for different statistical distances. Four classic statistics were included for analysis. In the future, other more sophisticated distances could be incorporated in the modified distance measure. The SLIC procedure-based algorithms, including SLIC, GC-SLIC, IER, and the proposed methods, were compared with Ncut and TurboPixel methods. Results show the effectiveness of the SLIC algorithms, despite simple and easy to implement. The present SLIC approach can also be extended to the multiphase image segmentation, which is usually used as a preparation step for the applications, such as region-based change detection.

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